

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

This document develops a three-generator sigma algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$ and signature $(+, -, -, -)$ are used.

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1 The algebra at a glance

The generators σ_n are algebraic objects with ordered multiplication rules. They are not ordinary vector components: their products generally do not commute, so the order of the factors is part of every expression. The following panel previews the generators, their product, and the general complex element developed below.

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations use

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3, \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \quad (1.2)$$

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (1.3)$$

Thus the square of a vector always denotes its self dot product.

The set $\{1, \boldsymbol{\sigma}, \mathbf{i}\boldsymbol{\sigma}, \mathbf{i}\}$ is a complete basis, displayed by type as

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

Anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}. \quad (1.4)$$

Thus its order of multiplication does not matter.

The commutator is

$$[X, Y] := XY - YX \quad (1.5)$$

$$[Y, X] = -[X, Y]. \quad (1.6)$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k \in \{1, 2, 3\}$, define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \quad (1.7)$$

and the Levi-Civita symbol by

$$\epsilon_{nmk} := \begin{cases} +1, & (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two or more indices are equal.} \end{cases} \quad (1.8)$$

The ordered sigma multiplication rule is then For $n, m \in \{1, 2, 3\}$,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9)$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k. \quad (1.10)$$

2 Hermitian and sigma-conjugate operators

Define the two operations by

$$(\cdot)^{\mathbf{H}} := \text{reverse the order of every product of sigma generators}, \quad (2.1)$$

$$(\cdot)^* := \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order}. \quad (2.2)$$

Thus, for two generators,

$$(\sigma_m \sigma_n)^{\mathbf{H}} = \sigma_n \sigma_m, \quad (2.3)$$

$$(\sigma_m \sigma_n)^* = (-\sigma_m)(-\sigma_n). \quad (2.4)$$

Both operations act term by term on sums. Their actions on the four basis types are

type representative	scalar 1	vector σ_n	bivector $\mathbf{i} \sigma_n$	pseudoscalar $\mathbf{i}$
$(\cdot)^*$	+	−	+	−
$(\cdot)^{\mathbf{H}}$	+	+	−	−
$(\cdot)^{* \mathbf{H}}$	+	−	−	+

In particular,

$$\sigma_n^* = -\sigma_n, \quad (2.5)$$

$$\sigma_n^{\mathbf{H}} = \sigma_n, \quad (2.6)$$

$$\mathbf{i}^* = -\mathbf{i}, \quad (2.7)$$

$$\mathbf{i}^{\mathbf{H}} = -\mathbf{i}. \quad (2.8)$$

For the bivectors, these actions combine to give

$$(\mathbf{i} \sigma_n)^* = \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n \quad (2.9)$$

$$(\mathbf{i} \sigma_n)^{\mathbf{H}} = \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n. \quad (2.10)$$

Thus sigma conjugation preserves the original order of the factors, whereas the Hermitian operator changes their order. For arbitrary algebra elements Z, Z' , these definitions give

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}}, \quad (ZZ')^* = Z^* Z'^*, \quad Z^{* \mathbf{H}} = Z^{\mathbf{H} *}.$$

Appendix A reproduces this as Hermitian conjugation in the Pauli-matrix representation.

3 Paravectors

3.1 Core operations

Let $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. First form the two real paravectors

$$X := a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.1)$$

$$Y := b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.2)$$

Their complex combination is the general complex paravector

$$Z := X + iY = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.3)$$

Equivalently, collect its complex scalar and vector coefficients as

$$S := a + bi \quad (3.4)$$

$$\mathbf{V} := \mathbf{A} + \mathbf{B}i \quad (3.5)$$

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.6)$$

The corresponding coefficient-extraction operations are

$$\text{sc}(Z) := S \quad (3.7)$$

$$\text{vec}(Z) := \mathbf{V}. \quad (3.8)$$

Sigma conjugation changes the sign of every sigma generator without changing multiplied order; the Hermitian operator changes multiplied sigma order. For the complex coefficients, write

$$S^{\text{H}} = S^* := a - bi \quad (3.9)$$

$$\mathbf{V}^{\text{H}} = \mathbf{V}^* := \mathbf{A} - \mathbf{B}i. \quad (3.10)$$

For a paravector, the two operations are then

$$Z^* := S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.11)$$

$$Z^{\text{H}} := S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{A} - \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.12)$$

Their composition is the adjugate:

$$\text{adj}(Z) := Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.13)$$

Define the paravector trace and determinant by

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2S, \quad (3.14)$$

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (3.15)$$

For any paravectors Z and Z' , the adjugate reverses product order:

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z). \quad (3.16)$$

Consequently, for $z^2 \neq 1$ where the inverse occurs,

$$\det(ZZ') = \text{adj}(ZZ')ZZ' \quad (3.17)$$

$$= \text{adj}(Z')(\text{adj}(Z) Z)Z' \quad (3.18)$$

$$= \det(Z) \det(Z'). \quad (3.19)$$

For any complex scalar c ,

$$\text{adj}(cZ) = c \text{adj}(Z). \quad (3.20)$$

Therefore

$$\det(cZ) = c^2 \det(Z). \quad (3.21)$$

Whenever $\det(Z) \neq 0$, define

$$Z^{-1} := \frac{\text{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2} \quad (3.22)$$

$$Z^{-1}Z = ZZ^{-1} = 1. \quad (3.23)$$

Differentiating $Z^{-1}Z = 1$ gives the general inverse rule

$$d(Z^{-1}) = -Z^{-1} dZ Z^{-1}. \quad (3.24)$$

Both trace and determinant are cyclic:

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (3.25)$$

$$\det(ZZ') = \det(Z'Z). \quad (3.26)$$

For three factors,

$$\text{tr}(ZZ'Z'') = \text{tr}(Z'Z''Z) = \text{tr}(Z''Z'Z'), \quad (3.27)$$

$$\det(ZZ'Z'') = \det(Z'Z''Z) = \det(Z''Z'Z'). \quad (3.28)$$

For any complex paravector Z and invertible complex paravector T , let

$$Z' := T^{-1}ZT. \quad (3.29)$$

Cyclicity then gives

$$\text{tr}(Z') = \text{tr}(T^{-1}ZT) = \text{tr}(ZTT^{-1}) = \text{tr}(Z), \quad (3.30)$$

$$\det(Z') = \det(T^{-1}ZT) = \det(ZTT^{-1}) = \det(Z). \quad (3.31)$$

Replacing T by T^{-1} gives the same two invariances for $Z' = TZT^{-1}$. Finally, define real and imaginary paravector extraction by

$$\text{re}(Z) := \frac{1}{2}(Z + Z^{\text{H}}) = X, \quad (3.32)$$

$$\text{im}(Z) := \frac{Z - Z^{\text{H}}}{2i} = Y. \quad (3.33)$$

The functions re and im also act componentwise on complex scalars and vectors. The scalar and vector parts are recovered by

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*\text{H}}), \quad (3.34)$$

$$\mathbf{V} = \text{vec}(Z), \quad (3.35)$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*\text{H}}). \quad (3.36)$$

The four real coefficients then follow by composition:

$$a = \text{re}(S) = \text{sc}(X) = \text{sc}(\text{re}(Z)), \quad (3.37)$$

$$b = \text{im}(S) = \text{sc}(Y) = \text{sc}(\text{im}(Z)), \quad (3.38)$$

$$\mathbf{A} = \text{re}(\mathbf{V}) = \text{vec}(X) = \text{vec}(\text{re}(Z)), \quad (3.39)$$

$$\mathbf{B} = \text{im}(\mathbf{V}) = \text{vec}(Y) = \text{vec}(\text{im}(Z)). \quad (3.40)$$

These definitions are intrinsic to the algebra. Appendix A later reproduces them through the faithful Pauli-matrix representation.

3.2 Products and quadratic forms

From the ordered sigma multiplication rule,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \quad (3.41)$$

$$= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} i \sigma_k \quad (3.42)$$

$$= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.43)$$

Adding the product with $\mathbf{A}$ and $\mathbf{B}$ interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}). \quad (3.44)$$

Consequently, for $z^2 \neq 1$ where the inverse occurs,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \quad (3.45)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \quad (3.46)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (3.47)$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}. \quad (3.48)$$

The corresponding commutator is

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.49)$$

Hence

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.50)$$

$$[\mathbf{A} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] = 2\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.51)$$

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.52)$$

3.3 Quadratic identity and eigenvalues

Since $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$ and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$,

$$(Z - S)^2 = \mathbf{V}^2, \quad (3.53)$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0. \quad (3.54)$$

Using $\text{tr}(Z) = 2S$ and $\det(Z) = S^2 - \mathbf{V}^2$ gives the equivalent trace-determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (3.55)$$

Replacing Z by a scalar eigenvalue λ and solving this quadratic gives

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}. \quad (3.56)$$

Because $\mathbf{V}$ is generally complex, $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0. \quad (3.57)$$

3.4 Real products

Let

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.58)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.59)$$

Here $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. Then

$$XY = (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.60)$$

$$\text{sc}(XY) = ab + \mathbf{A} \cdot \mathbf{B}, \quad (3.61)$$

$$\text{vec}(XY) = a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}, \quad (3.62)$$

$$[X, Y] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.63)$$

$$X^2 = (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \quad (3.64)$$

$$\text{sc}(X^2) = a^2 + \mathbf{A}^2, \quad (3.65)$$

$$\text{vec}(X^2) = 2a\mathbf{A}, \quad (3.66)$$

$$X^2 - 2aX + (a^2 - \mathbf{A}^2) = 0. \quad (3.67)$$

The condition $\mathbf{A} \times \mathbf{B} = 0$ implies $XY = YX$ and $X^{-1}Y = YX^{-1}$.

3.5 Complex products

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.68)$$

Thus

$$\text{sc}(ZZ') = SS' + \mathbf{V} \cdot \mathbf{V}', \quad (3.69)$$

$$\text{vec}(ZZ') = S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}'. \quad (3.70)$$

Equivalently, if $Z = X + \mathbf{i}Y$ and $Z' = X' + \mathbf{i}Y'$, then

$$ZZ' = (XX' - YY') + \mathbf{i}(XY' + YX'). \quad (3.71)$$

Let

$$Z = (a + \mathbf{b}\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, \quad (3.72)$$

where $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, and use primes for a second paravector with real primed components. The four real components of ZZ' are

$$\text{real scalar: } aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \quad (3.73)$$

$$\text{imaginary scalar: } ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \quad (3.74)$$

$$\text{real vector: } a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \quad (3.75)$$

$$\text{imaginary vector: } a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \quad (3.76)$$

For the important quadratic products:

$$Z^2 : (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.77)$$

$$ZZ^*, Z^*Z : (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) \mp 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.78)$$

$$ZZ^H, Z^H Z : a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} \pm \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.79)$$

$$Z \operatorname{adj}(Z) : a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (3.80)$$

The scalar part is cyclic, $\operatorname{sc}(ZZ') = \operatorname{sc}(Z'Z)$. The operator laws already established also give $(ZZ')^{*H} = Z'^{*H}Z^{*H}$. The paired signs follow the displayed product order. Thus $ZZ^* = Z^*Z$ and $ZZ^H = Z^H Z$ precisely when $\mathbf{A} \times \mathbf{B} = 0$.

Norms. For $S = a + bi$ and $\mathbf{V} = \mathbf{A} + \mathbf{B}i$, with real components $a, b, \mathbf{A}, \mathbf{B}$, define

$$\|S\|^2 := SS^* = S^*S = a^2 + b^2, \quad (3.81)$$

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} = \mathbf{A}^2 + \mathbf{B}^2, \quad (3.82)$$

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \quad (3.83)$$

Equivalently,

$$\|Z\|^2 = \operatorname{sc}(Z^H Z) = \frac{1}{2} \operatorname{tr}(Z^H Z) = \operatorname{sc}(ZZ^H) = \operatorname{sc}(Z^* Z^{*H}) = \operatorname{sc}(Z^{*H} Z^*). \quad (3.84)$$

These norms are not algebraic squares; in general,

$$S^2 \neq \|S\|^2 \quad (3.85)$$

$$\mathbf{V}^2 \neq \|\mathbf{V}\|^2. \quad (3.86)$$

For any complex scalar c ,

$$\|cZ\|^2 = \|c\|^2 \|Z\|^2. \quad (3.87)$$

Comparison with the determinant. The norm is real and nonnegative, whereas $\det(Z)$ may be complex. For $c = re^{i\theta}$ with real r and θ , the determinant retains the phase while the norm does not:

$$\det(re^{i\theta} Z) = r^2 \det(Z) e^{2i\theta}, \quad (3.88)$$

$$\|re^{i\theta} Z\|^2 = r^2 \|Z\|^2. \quad (3.89)$$

3.6 Projection operators

For a real unit vector $\mathbf{u}$, define

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.90)$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\Pi_{\pm}^2 = \Pi_{\pm}, \quad (3.91)$$

$$\Pi_+ \Pi_- = 0, \quad (3.92)$$

$$\Pi_+ + \Pi_- = 1, \quad (3.93)$$

$$\Pi_{\pm}^H = \Pi_{\pm}, \quad (3.94)$$

$$\Pi_{\pm}^* = \Pi_{\mp}, \quad (3.95)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} = \pm \Pi_{\pm}. \quad (3.96)$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For two unit directions,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.97)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.98)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.99)$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. Then

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \quad (3.100)$$

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \Pi_{\mp}. \quad (3.101)$$

For comparison, define $\Pi'_{\pm} := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})$. Their products are

$$\Pi_+\Pi'_+ = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} + \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.102)$$

$$\Pi_+\Pi'_- = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} - \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.103)$$

$$\Pi_-\Pi'_+ = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} + \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.104)$$

$$\Pi_-\Pi'_- = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} - \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.105)$$

3.7 Ladder operators

For projections along the third axis, define

$$Z_{\pm} = \frac{1}{2}(\sigma_1 \pm i\sigma_2) \quad (3.106)$$

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \sigma_3). \quad (3.107)$$

Then

$$[\sigma_3, Z_{\pm}] = \pm 2Z_{\pm}, \quad (3.108)$$

$$Z_{\pm}^2 = 0, \quad (3.109)$$

$$Z_{\pm}Z_{\mp} = \Pi_{\pm}, \quad (3.110)$$

$$\sigma_1 = Z_+ + Z_-, \quad (3.111)$$

$$\sigma_2 = (Z_+ - Z_-)/i, \quad (3.112)$$

$$\sigma_3 = \Pi_+ - \Pi_-. \quad (3.113)$$

The ladder operators map between the two projected sectors according to

$$Z_+\Pi_+ = 0, \quad (3.114)$$

$$Z_+\Pi_- = Z_+, \quad (3.115)$$

$$\Pi_+Z_+ = Z_+, \quad (3.116)$$

$$\Pi_-Z_+ = 0, \quad (3.117)$$

$$Z_-\Pi_- = 0, \quad (3.118)$$

$$Z_-\Pi_+ = Z_-, \quad (3.119)$$

$$\Pi_-Z_- = Z_-, \quad (3.120)$$

$$\Pi_+Z_- = 0. \quad (3.121)$$

3.8 Functions of paravectors

General projector rule. Let

$$\lambda_+, \lambda_- \in \mathbb{C}. \quad (3.122)$$

Suppose $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+ + \Pi_- = 1$, $\Pi_+ \Pi_- = \Pi_- \Pi_+ = 0$, and

$$Z = \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.123)$$

Then orthogonality gives

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_-. \quad (3.124)$$

Thus, for a convergent power series $f(w) = \sum_{n=0}^{\infty} c_n w^n$,

$$f(Z) = \sum_{n=0}^{\infty} c_n Z^n \quad (3.125)$$

$$= f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.126)$$

Functions of a unit vector. For $X = \mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$, the eigenvalues are ± 1 . The general rule and $\Pi_\pm = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$ give

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(1) \Pi_+ + f(-1) \Pi_- = \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.127)$$

Consequently,

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm \Pi_+ \pm i \Pi_-, \quad (3.128)$$

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2i\pi(n\Pi_+ + (\frac{1}{2} + m)\Pi_-), \quad (3.129)$$

$$(z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} = \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \quad (3.130)$$

$$\frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} = i \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.131)$$

$$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.132)$$

$$\exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cos \theta + i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.133)$$

For $0 < \|\mathbf{r}\| < 1$, take $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$. Then

$$\frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} = \exp(-2 \tanh^{-1}(\|\mathbf{r}\|)) \Pi_+ + \exp(2 \tanh^{-1}(\|\mathbf{r}\|)) \Pi_-. \quad (3.134)$$

Real paravectors. Let $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, where $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. For $\mathbf{A} \neq 0$, put $\mathbf{u} = \mathbf{A}/\|\mathbf{A}\|$ and define

$$\lambda_\pm := a \pm \|\mathbf{A}\|. \quad (3.135)$$

Using $1 = \Pi_+ + \Pi_-$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$ gives

$$X = a(\Pi_+ + \Pi_-) + \|\mathbf{A}\|(\Pi_+ - \Pi_-) \quad (3.136)$$

$$= \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.137)$$

The general rule therefore gives

$$f(X) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.138)$$

With nonzero eigenvalues where required and chosen function branches, this includes

$$X^n = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.139)$$

$$X^{-1} = \lambda_+^{-1}\Pi_+ + \lambda_-^{-1}\Pi_-, \quad (3.140)$$

$$\sqrt{X} = \pm\sqrt{\lambda_+}\Pi_+ \pm \sqrt{\lambda_-}\Pi_-, \quad (3.141)$$

$$\log X = \log(\lambda_+)\Pi_+ + \log(\lambda_-)\Pi_-, \quad (3.142)$$

$$\exp(X) = \exp(a) (\cosh\|\mathbf{A}\| + \sinh\|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (3.143)$$

$$\exp(i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos\|\mathbf{A}\| + i\sin\|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.144)$$

The real vector exponential has a direct kinematic interpretation. With $\theta = \|\mathbf{A}\|$, define

$$\gamma := \cosh \theta \quad (3.145)$$

$$\mathbf{v} := \tanh \theta \mathbf{u}. \quad (3.146)$$

Then $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ and

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (3.147)$$

Thus the magnitude of the exponent is the rapidity, while its direction is the direction of motion. This is the normalized future-directed paravector used below for four-velocity. The spectral inverse agrees with the adjugate–determinant formula developed earlier. Indeed,

$$\text{adj}(X) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.148)$$

$$\det(X) = \lambda_+ \lambda_-, \quad (3.149)$$

and therefore, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(X)}{\det(X)} = \lambda_+^{-1}\Pi_+ + \lambda_-^{-1}\Pi_- = X^{-1}. \quad (3.150)$$

The formulas extend continuously to $\mathbf{A} = 0$.

For direct component formulas, if $X^2 = Y$, then

$$a^2 = \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right) \quad (3.151)$$

$$\mathbf{A}^2 = b - a^2, \quad (3.152)$$

with signs and vector direction chosen consistently. Conversely, where $b^2 > \mathbf{B}^2$, $b > 0$, and $\mathbf{B} \neq 0$,

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1} \left(\frac{\|\mathbf{B}\|}{b} \right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.153)$$

Complex paravectors. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$, let $z := \sqrt{\mathbf{V}^2} \in \mathbb{C}$ and define

$$\lambda_{\pm} := S \pm z. \quad (3.154)$$

When $z \neq 0$, define

$$\Pi_{\pm} := \frac{1}{2} \left(1 \pm \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z} \right). \quad (3.155)$$

Because $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2 = z^2$,

$$\Pi_{\pm}^2 = \frac{1}{4} \left(1 \pm 2 \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z} + \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^2}{z^2} \right) = \Pi_{\pm}, \quad (3.156)$$

$$\Pi_+ \Pi_- = \frac{1}{4} \left(1 - \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^2}{z^2} \right) = 0. \quad (3.157)$$

Also, $1 = \Pi_+ + \Pi_-$ and $\mathbf{V} \cdot \boldsymbol{\sigma} = z(\Pi_+ - \Pi_-)$, so

$$Z = S(\Pi_+ + \Pi_-) + z(\Pi_+ - \Pi_-) \quad (3.158)$$

$$= \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.159)$$

The general rule gives

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.160)$$

The adjugate and determinant have the corresponding spectral forms

$$\text{adj}(Z) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.161)$$

$$\det(Z) = \lambda_+ \lambda_-. \quad (3.162)$$

Consequently, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_- = Z^{-1}. \quad (3.163)$$

Choosing square-root branches gives

$$\sqrt{Z} = \sqrt{\lambda_+} \Pi_+ + \sqrt{\lambda_-} \Pi_- \quad (3.164)$$

$$= \frac{1}{2}(\sqrt{\lambda_+} + \sqrt{\lambda_-}) + \frac{\sqrt{\lambda_+} - \sqrt{\lambda_-}}{2z} \mathbf{V} \cdot \boldsymbol{\sigma}, \quad (3.165)$$

and, when $\lambda_{\pm} \neq 0$, choosing logarithm branches gives

$$\log Z = \log(\lambda_+) \Pi_+ + \log(\lambda_-) \Pi_- \quad (3.166)$$

$$= \frac{1}{2}(\log(\lambda_+) + \log(\lambda_-)) + \frac{\log(\lambda_+) - \log(\lambda_-)}{2z} \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.167)$$

Also,

$$\exp(Z) = \exp(S) \left(\cosh z + \frac{\sinh z}{z} \mathbf{V} \cdot \boldsymbol{\sigma} \right), \quad (3.168)$$

$$\exp(Z^*) = \exp(Z)^*, \quad (3.169)$$

$$\exp(Z^H) = \exp(Z)^H, \quad (3.170)$$

$$\exp(-Z) = \exp(Z)^{-1}. \quad (3.171)$$

At $z = 0$, these formulas are understood by analytic limits whenever the chosen function branch is analytic near S .

3.9 Differentiated fields

For products of fields,

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (3.172)$$

Let $F = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$ depend on $(t, \mathbf{r})$. Then

$$\partial F = (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \quad (3.173)$$

$$+ (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \quad (3.174)$$

$$+ i(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.175)$$

Multiplication by i rotates these four components: the real scalar becomes $-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$, the imaginary scalar becomes $\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real vector becomes the negative of the imaginary vector above, and the imaginary vector becomes the real vector.

4 Transformations and invariants

The transformations used below are organized by the quantities they leave unchanged:

$$\text{tr}(Z) = 2 \text{sc}(Z) \quad (4.1)$$

$$\det(Z) \quad (4.2)$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(Z^H Z). \quad (4.3)$$

The trace is linear; the determinant and norm are quadratic. Here T and T' denote general complex paravectors, with T' a second transformation factor rather than a derivative. The symbol U denotes a complex unitary vector:

$$U^H U = U U^H = 1 \quad (4.4)$$

$$U^H = U^{-1}. \quad (4.5)$$

Here U is a transformation; Section 5 later reuses the symbol for four-velocity.

Trace invariance. For invertible T , consider either similarity:

$$Z' = T^{-1} Z T, \quad (4.6)$$

$$Z' = T Z T^{-1}. \quad (4.7)$$

For example, cyclicity gives

$$\text{tr}(T^{-1} Z T) = \text{tr}(Z T T^{-1}) = \text{tr}(Z). \quad (4.8)$$

Thus $\text{tr}(Z') = \text{tr}(Z)$. For unitary U , the same similarities are

$$Z' = U^H Z U, \quad (4.9)$$

$$Z' = U Z U^H. \quad (4.10)$$

Determinant invariance. For the general two-sided transformation $Z' = T'ZT$,

$$\det(Z') = \det(T') \det(T) \det(Z). \quad (4.11)$$

Hence $\det(T') \det(T) = 1$ guarantees $\det(Z') = \det(Z)$, and is required if this equality is to hold for arbitrary Z . Equivalently, when $\det(T) \neq 0$, any T' satisfying

$$\det(T') = \frac{1}{\det(T)} \quad (4.12)$$

preserves the determinant of every Z . The automatic choice $T' = T^{-1}$ gives the similarity

$$Z' = T^{-1}ZT. \quad (4.13)$$

The opposite ordering $Z' = TZT^{-1}$ preserves the determinant as well. For unitary U ,

$$\det(U^H) \det(U) = \det(U^{-1}) \det(U) = 1, \quad (4.14)$$

so both unitary sandwiches preserve the determinant.

Norm invariance. Norm invariance requires the unitary condition (4.4). For example, if $Z' = U^H Z U$, then

$$\|Z'\|^2 = \frac{1}{2} \text{tr}(U^H Z^H U U^H Z U) \quad (4.15)$$

$$= \frac{1}{2} \text{tr}(Z^H Z) = \|Z\|^2. \quad (4.16)$$

The same equality $\|Z'\|^2 = \|Z\|^2$ holds for

$$Z' = UZ \quad (4.17)$$

$$Z' = ZU \quad (4.18)$$

$$Z' = U^H Z U \quad (4.19)$$

$$Z' = UZU^H. \quad (4.20)$$

It does not hold in general when T is nonunitary.

Rotations and boosts preserve different combinations of these invariants. A rotation satisfies

$$R^H = R^{-1} \quad (4.21)$$

$$R^H R = R R^H = 1, \quad (4.22)$$

so R is unitary; $Z' = R^H Z R$ preserves trace, determinant, and norm. A nonzero boost satisfies

$$L^H = L \quad (4.23)$$

$$\det(L) = 1 \quad (4.24)$$

$$L^H L = L^2 \neq 1, \quad (4.25)$$

so L is not unitary. The transformation $Z' = L Z L$ preserves the determinant but does not generally preserve the trace or norm.

5 Space–time kinematics

5.1 Intervals, velocity, and momentum

Represent a differential event by the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (5.1)$$

$$dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (5.2)$$

The Minkowski interval is

$$ds^2 = \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2. \quad (5.3)$$

For any invertible complex paravector T , define

$$dX' := T^{-1} dX T. \quad (5.4)$$

The cyclic determinant property gives

$$\det(dX') = \det(T^{-1} dX T) = \det(dX T T^{-1}) = \det(dX) = ds^2. \quad (5.5)$$

Thus similarity by any such T leaves the determinant unchanged. This is an algebraic change of basis; even when $T = L$, it is not the physical boost of a space–time paravector. The latter is the congruence $dX' = L dX L$ developed in Section 8.2. The alternative similarity ordering $T dX T^{-1}$ has the same determinant invariance. For a future-directed time-like interval, choose the positive scalar root

$$ds := \sqrt{\det(dX)}. \quad (5.6)$$

The sign of the invariant squared interval classifies the causal relation between the two neighboring events:

$$ds^2 > 0 : \quad \text{time-like; a massive particle can follow the interval,} \quad (5.7)$$

$$ds^2 = 0 : \quad \text{null; light can follow the interval,} \quad (5.8)$$

$$ds^2 < 0 : \quad \text{space-like; no causal signal can follow the interval.} \quad (5.9)$$

Only in the time-like case is ds real and nonzero under the chosen positive-root convention; along a time-like worldline it is the proper-time differential. Thus the following kinematics explores the determinant of dX and its normalization by this scalar. For two intervals,

$$[dX, dX'] = 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (5.10)$$

When $ds^2 \neq 0$,

$$dX^{-1} = \frac{\text{adj}(dX)}{\det(dX)} = \frac{dX^*}{ds^2}. \quad (5.11)$$

A null interval has $dt^2 = d\mathbf{r}^2$ and no inverse. Along a time-like worldline, s is proper time. Since $d\mathbf{r} = \mathbf{v} dt$,

$$ds^2 = (1 - \mathbf{v}^2) dt^2 \quad (5.12)$$

$$ds = \frac{dt}{\gamma} \quad (5.13)$$

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}}. \quad (5.14)$$

Hence

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \quad (5.15)$$

$$\gamma = \frac{dt}{ds} \quad (5.16)$$

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (5.17)$$

Using the scalar-scaling rule (3.21), its determinant is

$$\det(U) = \det\left(\frac{dX}{ds}\right) = \frac{\det(dX)}{ds^2} = 1. \quad (5.18)$$

Equivalently, $\det(U) = \gamma^2(1 - \mathbf{v}^2) = 1$, so

$$U^{-1} = \frac{\text{adj}(U)}{\det(U)} = U^*. \quad (5.19)$$

Proper time therefore obeys

$$\tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt = \int \frac{dt}{\gamma}. \quad (5.20)$$

The four-momentum is the paravector

$$mU = m\gamma + m\gamma\mathbf{v} \cdot \boldsymbol{\sigma} = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (5.21)$$

Its scalar and vector components define

$$E := m\gamma \quad (5.22)$$

$$\mathbf{P} := m\gamma\mathbf{v}. \quad (5.23)$$

Taking its determinant gives the mass-shell relation

$$\det(mU) = m^2 \det(U) = m^2 = \det(E + \mathbf{P} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{P}^2, \quad (5.24)$$

or $E^2 = \mathbf{P}^2 + m^2$; for $m = 0$, $E^2 = \mathbf{P}^2$.

For coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$,

$$\frac{d\gamma}{dt} = \gamma^3 \mathbf{v} \cdot \mathbf{a} \quad (5.25)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}, \quad (5.26)$$

so

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}, \quad (5.27)$$

Differentiating $U^*U = 1$ gives

$$\frac{dU^*}{ds}U + U^*\frac{dU}{ds} = 0 \quad (5.28)$$

$$\text{sc}(U^* dU/ds) = 0. \quad (5.29)$$

If $A = dU/ds$, then

$$\det(A) = -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2). \quad (5.30)$$

For $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$, define $\mathbf{a}_{\parallel} = (\mathbf{a} \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$. Then

$$\det(A) = -\gamma^4(\mathbf{a}_{\perp}^2 + \gamma^2\mathbf{a}_{\parallel}^2). \quad (5.31)$$

The product U^*A has zero scalar part, real vector $\gamma^3\mathbf{a}$, and imaginary vector $-\gamma^3\mathbf{v} \times \mathbf{a}$; while UA has real scalar $2\gamma^5\mathbf{v} \cdot \mathbf{a}$, real vector $\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$, and imaginary vector $\gamma^3\mathbf{v} \times \mathbf{a}$.

5.2 Varying transformations and second differentials

Along a path, let d denote successive differentiation with respect to its parameter and define

$$dX' = T^{-1} dX T. \quad (5.32)$$

Although $\det(dX') = \det(dX)$ at each point, a varying T adds a term to the second differential. Define

$$d\Theta := T^{-1} dT \quad (5.33)$$

$$dT^{-1} = -d\Theta T^{-1}. \quad (5.34)$$

Direct differentiation gives

$$d^2 X' = d(T^{-1} dX T) \quad (5.35)$$

$$= T^{-1} d^2 X T - d\Theta dX' + dX' d\Theta \quad (5.36)$$

$$= T^{-1} d^2 X T + [dX', d\Theta]. \quad (5.37)$$

Dividing by the corresponding powers of ds , define

$$\Omega := \frac{d\Theta}{ds} = T^{-1} \frac{dT}{ds}, \quad (5.38)$$

so, with $U = dX/ds$ and $U' = dX'/ds$,

$$\frac{dU'}{ds} = T^{-1} \frac{dU}{ds} T + [U', \Omega]. \quad (5.39)$$

For constant T , the connection term vanishes. For varying T , it is the algebraic source of the additional terms in rotating and accelerating frames and is the pathwise prototype of a covariant derivative.

Here dX' is defined by the pointwise transformation of the tangent. When T varies, it is not generally $d(T^{-1}XT)$, which also contains the transformation of X itself. Moreover, a general complex T can make dX' complex even when dX is real, so a physical space-time transformation requires an additional reality-preserving condition.

The corresponding local algebra basis is

$$\sigma'_n := T^{-1} \sigma_n T. \quad (5.40)$$

Similarity preserves all products, so the transformed generators obey the same sigma algebra. If T varies along the path, direct differentiation gives

$$d\sigma'_n = [\sigma'_n, d\Theta] \quad (5.41)$$

$$d\Theta = T^{-1} dT. \quad (5.42)$$

This equation describes the change of a local algebra basis. It should not be identified with a physical Lorentz transformation until the action of T and the condition preserving real space-time paravectors have both been specified. In particular, the complex boost-like terms obtained from a completely general T are connection coefficients, not automatically components of a real physical velocity.

5.3 Free-particle action

The free-particle action and Lagrangian are

$$\int L dt = -m \int ds \quad (5.43)$$

$$L = -m\sqrt{1 - \mathbf{v}^2} = -\frac{m}{\gamma}. \quad (5.44)$$

For a general Lagrangian $L(t, \mathbf{r}, \mathbf{v})$,

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L \quad (5.45)$$

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (5.46)$$

$$\frac{dH}{dt} = -\partial_t L. \quad (5.47)$$

Thus spatial translation invariance conserves canonical momentum and time translation invariance conserves the Hamiltonian. For the free-particle Lagrangian,

$$\partial_{\mathbf{v}} L = m\gamma \mathbf{v}, \quad (5.48)$$

$$H = m\gamma \mathbf{v}^2 + \frac{m}{\gamma} = m\gamma(\mathbf{v}^2 + \gamma^{-2}) = m\gamma. \quad (5.49)$$

At low velocity, the binomial expansions are

$$\sqrt{1 - \mathbf{v}^2} = 1 - \frac{1}{2}\mathbf{v}^2 + O(v^4) \quad (5.50)$$

$$\gamma = 1 + \frac{1}{2}\mathbf{v}^2 + O(v^4). \quad (5.51)$$

Multiplying the first expansion by $-m$ gives

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m + \frac{1}{2}m\mathbf{v}^2 + O(v^4), \quad (5.52)$$

while the second gives

$$H = m\gamma = m + \frac{1}{2}m\mathbf{v}^2 + O(v^4). \quad (5.53)$$

The exact relativistic kinetic energy is therefore

$$K := H - m = m(\gamma - 1), \quad (5.54)$$

and, apart from the constant rest-energy term, both expansions recover the nonrelativistic kinetic energy $\frac{1}{2}m\mathbf{v}^2$.

6 Electromagnetism

6.1 Field equation and potentials

Introduce the paravector differential operator, electromagnetic field, and source

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \quad (6.1)$$

$$F = (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma} \quad (6.2)$$

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.3)$$

All four Maxwell equations are the single equation

$$\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.4)$$

Sigma conjugation gives the equivalent equation

$$\partial^* F^* = D. \quad (6.5)$$

Expanding the left side by component gives

$$\partial F = \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i \partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \quad (6.6)$$

$$+ \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.7)$$

$$+ \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.8)$$

Equating these four components with those of $\rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives, respectively,

$$\text{real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (6.9)$$

$$\text{imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (6.10)$$

$$\text{real vector: } \partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad (6.11)$$

$$\text{imaginary vector: } \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (6.12)$$

The conjugate operator products give

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 =: \square, \quad (6.13)$$

whereas

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}}^2 + 2\partial_t \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma} \neq \square. \quad (6.14)$$

Applying ∂^* to (6.4) gives

$$\square F = \partial^* D^* \quad (6.15)$$

$$= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (6.16)$$

The real-scalar component is charge conservation, and the real- and imaginary-vector components are the electric and magnetic wave equations:

$$\text{real scalar: } \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (6.17)$$

$$\text{real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (6.18)$$

$$\text{imaginary vector: } \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (6.19)$$

In vacuum, $\rho = 0$ and $\mathbf{J} = 0$, so

$$\square \mathbf{E} = 0 \quad (6.20)$$

$$\square \mathbf{B} = 0. \quad (6.21)$$

For the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$,

$$\partial\Phi = \underbrace{\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}}_{\text{real scalar}} + \underbrace{(\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.22)$$

$$+ \underbrace{i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.23)$$

Define the scalar component

$$S := \text{sc}(\partial\Phi) = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (6.24)$$

Then $\partial\Phi = S + F^*$: its real-vector and imaginary-vector components give

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V \quad (6.25)$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (6.26)$$

Also, $F = \partial^* \Phi^* - S$, so

$$\partial F = \partial \partial^* \Phi^* - \partial S = \square \Phi^* - \partial S \quad (6.27)$$

$$= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}. \quad (6.28)$$

Comparing its real-scalar and real-vector components with $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives

$$\rho = \square V - \partial_t S \quad (6.29)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (6.30)$$

For a scalar field λ , the gauge transformation is

$$\Phi' = \Phi + \partial^* \lambda, \quad (6.31)$$

$$\partial\Phi' = \partial\Phi + \partial\partial^* \lambda = \partial\Phi + \square\lambda. \quad (6.32)$$

Because $\square\lambda$ is scalar, only the scalar component changes:

$$S' = S + \square\lambda \quad (6.33)$$

$$\text{vec}(\partial\Phi') = \text{vec}(\partial\Phi). \quad (6.34)$$

Thus F is unchanged. In the Lorenz gauge $S = 0$, equation (6.29) reduces to

$$\rho = \square V \quad (6.35)$$

$$\mathbf{J} = \square \mathbf{A}. \quad (6.36)$$

6.2 Field invariants

Electric and magnetic fields mix under Lorentz transformations, so their individual magnitudes depend on the observer. Two combinations remain the same for every inertial observer and therefore classify the electromagnetic field itself.

For

$$F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (6.37)$$

the scalar part vanishes, and therefore $\text{adj}(F) = -F$. Hence

$$\det(F) = -F^2 \quad (6.38)$$

$$= -(\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot (\mathbf{E} + \mathbf{B}\mathbf{i}) \quad (6.39)$$

$$= -\mathbf{E}^2 + \mathbf{B}^2 - 2\mathbf{i}\mathbf{E} \cdot \mathbf{B}. \quad (6.40)$$

For any invertible real or complex paravector T , let

$$F' = TFT^{-1}. \quad (6.41)$$

The general similarity identity (3.31) gives

$$\det(F') = \det(TFT^{-1}) = \det(F). \quad (6.42)$$

Rotations and boosts are the physical special cases

$$F' = RFR^{-1} = RFR^{\mathbf{H}} \quad (6.43)$$

$$F' = LFL^{-1} = LFL^*, \quad (6.44)$$

because $R^{-1} = R^{\mathbf{H}}$ and $L^{-1} = L^*$. Hence every inertial observer agrees on

$$a = \mathbf{E}^2 - \mathbf{B}^2 = -\text{re}(\det(F)), \quad (6.45)$$

$$b = \mathbf{E} \cdot \mathbf{B} = -\frac{1}{2}\text{im}(\det(F)). \quad (6.46)$$

For nonzero fields, these invariants give the standard classification:

$$b = 0 : \quad \text{electric-like when } a \text{ is positive; some frame has no magnetic field,} \quad (6.47)$$

$$b = 0 : \quad \text{magnetic-like when } a \text{ is negative; some frame has no electric field,} \quad (6.48)$$

$$b \neq 0 : \quad \text{no frame can eliminate either field.} \quad (6.49)$$

When both a and b vanish, the field is null or radiation-like: the electric and magnetic fields are orthogonal and have equal magnitude. Thus the invariants reveal physical properties that cannot be inferred from one observer's values of $\mathbf{E}$ and $\mathbf{B}$ alone. Unlike the energy density and flux below, they do not change with the observer.

6.3 Lorentz force from the field and action

The Lorentz proper-force equation is

$$m \frac{dU}{ds} = q \text{re}(FU) = \frac{q}{2}(FU + UF^{\mathbf{H}}). \quad (6.50)$$

The Hermitian extraction removes the imaginary magnetic-dual components. The remaining scalar and vector components are, respectively,

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E} \quad (6.51)$$

$$\frac{d\mathbf{P}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (6.52)$$

The same result follows from the potential through the interacting-particle action

$$\int L \, dt = -m \int ds - q \int \text{sc}(\Phi^* U) \, ds \quad (6.53)$$

$$= \int \left(-\frac{m}{\gamma} - q \text{sc}(\Phi^* U/\gamma) \right) dt \quad (6.54)$$

$$= \int \left(-\frac{m}{\gamma} - q \text{sc}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right) dt \quad (6.55)$$

$$= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \quad (6.56)$$

Thus

$$L = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}) \quad (6.57)$$

$$\partial_{\mathbf{v}} L = m\gamma \mathbf{v} + q\mathbf{A} = \mathbf{P} + q\mathbf{A}. \quad (6.58)$$

The Euler–Lagrange terms are

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \frac{d\mathbf{P}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \quad (6.59)$$

$$\partial_{\mathbf{r}} L = -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \quad (6.60)$$

Using $(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives

$$\frac{d\mathbf{P}}{dt} = -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A} + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}) \quad (6.61)$$

$$= q(-\partial_{\mathbf{r}} V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})) \quad (6.62)$$

$$= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (6.63)$$

The Hamiltonian follows with the gauge-dependent terms cancelling:

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (6.64)$$

$$= m\gamma \mathbf{v}^2 + q\mathbf{v} \cdot \mathbf{A} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \quad (6.65)$$

$$= m\gamma(\mathbf{v}^2 + \gamma^{-2}) + qV = m\gamma + qV. \quad (6.66)$$

Finally, $dH/dt = -\partial_t L$ gives

$$\frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \quad (6.67)$$

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot (-\partial_{\mathbf{r}} V - \partial_t \mathbf{A}) = q\mathbf{v} \cdot \mathbf{E}. \quad (6.68)$$

To see equation (6.50) componentwise, expand

$$\frac{FU}{\gamma} = \mathbf{v} \cdot \mathbf{E} + i\mathbf{v} \cdot \mathbf{B} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + i(\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.69)$$

Taking the real paravector part removes the two imaginary components.

6.4 Energy balance

Since

$$F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma} \quad (6.70)$$

$$F^H = (\mathbf{E} - \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (6.71)$$

their ordered product is

$$FF^H = \mathbf{E}^2 + \mathbf{B}^2 + 2(\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (6.72)$$

Therefore the standard electromagnetic energy density u and Poynting vector $\mathbf{S}$ are the scalar and vector coefficients of $\frac{1}{2}FF^H$:

$$u := \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) = \frac{1}{2} \text{sc}(FF^H), \quad (6.73)$$

$$\mathbf{S} := \mathbf{E} \times \mathbf{B} = \frac{1}{2} \text{vec}(FF^H). \quad (6.74)$$

These quantities describe energy measured in the chosen frame; they are not the two Lorentz invariants contained in F^2 . The product $\frac{1}{2}FF^H$ packages energy density and energy flux into a paravector, but is not by itself the full electromagnetic stress-energy tensor. This product order gives the conventional sign of the Poynting vector directly. For comparison, reversing the order preserves the scalar coefficient and changes the sign of the vector coefficient:

$$\text{sc}(F^H F) = \text{sc}(FF^H) \quad (6.75)$$

$$\text{vec}(F^H F) = -\text{vec}(FF^H). \quad (6.76)$$

Starting only from the paravector field equation $\partial F = D^*$, its real-vector and imaginary-vector components give

$$\partial_t \mathbf{E} = \partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J} \quad (6.77)$$

$$\partial_t \mathbf{B} = -\partial_{\mathbf{r}} \times \mathbf{E}. \quad (6.78)$$

Taking their dot products with $\mathbf{E}$ and $\mathbf{B}$, respectively, yields

$$\partial_t u = \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} \quad (6.79)$$

$$= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B}) - \mathbf{E} \cdot \mathbf{J} - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \quad (6.80)$$

$$= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \quad (6.81)$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Substitution into $\partial_t u$ therefore derives Poynting's theorem, the local electromagnetic energy-balance law:

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (6.82)$$

Here $\mathbf{J} \cdot \mathbf{E}$ is the power density transferred from the field to matter; the minus sign is the corresponding loss of field energy. Equivalently, in paravector coefficients,

$$\frac{1}{2} \partial_t \text{sc}(FF^H) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(FF^H) = -\mathbf{J} \cdot \mathbf{E}. \quad (6.83)$$

7 Relativistic wave equations

7.1 Paravector inner products and expectations

For paravector amplitudes ψ, ψ' and a paravector Z , define the inner product and matrix element intrinsically by scalar extraction:

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^{\text{H}} \psi), \quad (7.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^{\text{H}} Z \psi). \quad (7.2)$$

The first is the $Z = 1$ case of the second, and $\langle \psi | \psi \rangle = \|\psi\|^2$. Thus normalization is $\text{sc}(\psi^{\text{H}} \psi) = 1$, not $\psi^{\text{H}} \psi = 1$. For an observable $Z^{\text{H}} = Z$ acting from the left,

$$\langle Z \rangle_{\psi} := \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^{\text{H}} Z \psi)}{\text{sc}(\psi^{\text{H}} \psi)}. \quad (7.3)$$

For normalized ψ , this reduces to $\langle Z \rangle_{\psi} = \langle \psi | Z | \psi \rangle$. Appendix A.4 shows that these intrinsic definitions become the usual trace pairing and, after minimal-ideal projection, the standard two-component column-spinor formulas.

For a paravector T with $\psi' = T\psi$,

$$\langle Z \rangle_{\psi'} = \frac{\text{sc}(\psi^{\text{H}} T^{\text{H}} Z T \psi)}{\text{sc}(\psi^{\text{H}} T^{\text{H}} T \psi)}. \quad (7.4)$$

If $T = U$, $U^{\text{H}} U = 1$, and $\langle \psi | \psi \rangle = 1$, then the denominator is unity and

$$\langle Z \rangle_{\psi'} = \text{sc}(\psi^{\text{H}} U^{\text{H}} Z U \psi). \quad (7.5)$$

Thus a unitary rotation preserves the normalization. A finite boost is not unitary, so this positive equal-time norm is not itself Lorentz invariant; the covariant Dirac normalization instead comes from the conserved Dirac current on a spacelike hypersurface.

7.2 Physical spinors and minimal ideals

A physical two-component Pauli spinor is represented intrinsically by a minimal ideal obtained by left projection in the paravector algebra. Choose a fixed unit direction $\mathbf{u}$ and use the projectors from Section 3.6,

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (7.6)$$

They satisfy

$$\Pi_{\pm}^2 = \Pi_{\pm}, \quad (7.7)$$

$$\Pi_+ \Pi_- = 0, \quad (7.8)$$

$$\Pi_+ + \Pi_- = 1, \quad (7.9)$$

$$\Pi_{\pm}^{\text{H}} = \Pi_{\pm}, \quad (7.10)$$

$$\Pi_{\pm}^* = \Pi_{\mp}, \quad (7.11)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} = \pm \Pi_{\pm}, \quad (7.12)$$

and hence $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$.

Project an unrestricted paravector amplitude from the left into either sector:

$$\psi'_\pm := \Pi_\pm \psi. \quad (7.13)$$

Then $\Pi_\pm \psi'_\pm = \psi'_\pm$ and $\Pi_\mp \psi'_\pm = 0$. Hermitian conjugation reverses the multiplied order, so the projected norm and observable matrix element are

$$\langle \psi'_\pm | \psi'_\pm \rangle = \text{sc}(\psi^H \Pi_\pm^H \Pi_\pm \psi) = \langle \psi | \Pi_\pm | \psi \rangle, \quad (7.14)$$

$$\langle \psi'_\pm | Z | \psi'_\pm \rangle = \text{sc}(\psi^H \Pi_\pm^H Z \Pi_\pm \psi) = \langle \psi | \Pi_\pm Z \Pi_\pm | \psi \rangle. \quad (7.15)$$

Appendix A.4 develops the faithful matrix image of this ideal and shows how it becomes one conventional two-component Pauli column. Applied to both paravector components of the Dirac block below, the same left projections select the conventional four-component Dirac sectors.

7.3 Klein–Gordon equation

With $E = i\partial_t$ and $\mathbf{P} = -i\partial_r$, the mass-shell relation becomes the scalar wave equation

$$(\square + m^2)\Psi = 0. \quad (7.16)$$

Writing $\Omega = \sqrt{m^2 - \partial_r^2}$ gives the frequency decomposition

$$(\partial_t \pm i\Omega)\Psi_\pm = 0 \quad (7.17)$$

$$\Psi = \Psi_+ + \Psi_-. \quad (7.18)$$

The square-root operator is understood spectrally: on a plane wave with wave vector $\mathbf{k}$, Ω multiplies by $\sqrt{m^2 + \mathbf{k}^2}$.

Minimal electromagnetic coupling replaces the operators by

$$E = i\partial_t - qV \quad (7.19)$$

$$\mathbf{P} = -i\partial_r - q\mathbf{A}, \quad (7.20)$$

so that $(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$.

7.4 Dirac equation in block form

For a paravector Z , define

$$\mathbf{W}(Z) := \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}. \quad (7.21)$$

It satisfies

$$\mathbf{W}(Z)^2 = \text{diag}(Z \text{adj}(Z), \text{adj}(Z) Z) \quad (7.22)$$

$$\mathbf{W}(Z)^* = \mathbf{W}(Z^*). \quad (7.23)$$

Since $\mathbf{W}(i\partial)^2 = -\square \mathbf{I}$, the first-order equation

$$(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0 \quad (7.24)$$

squares to the Klein–Gordon equation. For $\Psi = \{\Psi_1, \Psi_2\}^\top$ it reads

$$i\partial\Psi_2 - m\Psi_1 = 0 \quad (7.25)$$

$$i\partial^*\Psi_1 - m\Psi_2 = 0. \quad (7.26)$$

Conversely, for $m \neq 0$, any scalar solution Ψ' of $(\square + m^2)\Psi' = 0$ gives a Dirac solution

$$\Psi = \left\{ \Psi', \frac{i}{m}\partial^*\Psi' \right\}^\top. \quad (7.27)$$

Indeed, the second component equation holds by definition, and the first reduces to $-(\square + m^2)\Psi'/m = 0$. More generally, for constant complex scalars a and b , the same scalar solution gives

$$\Psi = \frac{1}{m} \left\{ (ma + ib\partial)\Psi', (ia\partial^* + mb)\Psi' \right\}^\top. \quad (7.28)$$

Substitution into the two component equations leaves respectively $-a(\square + m^2)\Psi'/m$ and $-b(\square + m^2)\Psi'/m$, so both vanish. This is a two-constant family generated from one Klein–Gordon solution, not a claim that one scalar solution parametrizes every Dirac solution. The characteristic equation is $(s + m)^2 + \square = 0$, hence $s = -m \pm i\sqrt{\square}$. With a potential, the gauge-covariant form is

$$(W(i\partial - q\Phi^*) - mI)\Psi = 0. \quad (7.29)$$

It is invariant under $\Phi' = \Phi + \partial^*\lambda$ and $\Psi = \exp(iq\lambda)\Psi'$.

7.5 Schrödinger–Pauli limit

Removing the rest-energy oscillation and expanding at low momentum gives the Pauli equation

$$i\partial_t\Psi = \left(\frac{(-i\partial_r - qA)^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi. \quad (7.30)$$

Using these projectors, $\Psi = \Psi_+ + \Psi_-$ with $\Psi_\pm = \Pi_\pm\Psi$. For a uniform static magnetic field, $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$, the sectors decouple:

$$i\partial_t\Psi_\pm = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi_\pm. \quad (7.31)$$

For a varying field direction the sectors are coupled. With H denoting the Pauli Hamiltonian, direct differentiation gives

$$i\partial_t\Psi_\pm = i(\partial_t\Pi_\pm)\Psi + \Pi_\pm H\Psi \quad (7.32)$$

$$= H\Psi_\pm + (i\partial_t\Pi_\pm + [\Pi_\pm, H])\Psi. \quad (7.33)$$

Because the field term commutes with its own instantaneous projectors, this is

$$i\partial_t\Psi_\pm = H\Psi_\pm + \mathbf{S}_\pm \quad (7.34)$$

$$\mathbf{S}_\pm = \left(i\partial_t\Pi_\pm + \frac{[\mathbf{P}^2, \Pi_\pm]}{2m} \right) \Psi. \quad (7.35)$$

7.6 Interacting operators and factorization

The covariant energy and momentum operators do not commute. Direct application to a test wavefunction gives

$$[E, P_m] = iqE_m \quad (7.36)$$

$$[P_m, P_n] = iq(\partial_m A_n - \partial_n A_m) \quad (7.37)$$

$$\mathbf{P} \times \mathbf{P} = iq\mathbf{B}. \quad (7.38)$$

Therefore

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF, \quad (7.39)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF^*, \quad (7.40)$$

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial^* - q\Phi, \quad (7.41)$$

$$E - \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial - q\Phi^*. \quad (7.42)$$

Expanding the minimally coupled Klein–Gordon operator gives the fully explicit result

$$(\square + iqS - q^2 \det(\Phi) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})) \Psi = 0. \quad (7.43)$$

The two orderings satisfy

$$0 = ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2) \Psi_1 = (-E^2 + \mathbf{P}^2 + m^2 + iqF^*) \Psi_1, \quad (7.44)$$

$$0 = ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2) \Psi_2 = (-E^2 + \mathbf{P}^2 + m^2 + iqF) \Psi_2. \quad (7.45)$$

These follow by multiplying the Dirac operator by its mass conjugate. In detail, define

$$\mathbf{D} := \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}, \quad (7.46)$$

$$\mathbf{D}' := \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}. \quad (7.47)$$

Then

$$\mathbf{D}'\mathbf{D} = -\text{diag}((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2, (\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2). \quad (7.48)$$

For a paravector wavefunction $\psi = \psi_s + \boldsymbol{\psi}_v \cdot \boldsymbol{\sigma}$,

$$\partial\psi = (\partial_t\psi_s + \partial_{\mathbf{r}} \cdot \boldsymbol{\psi}_v) + (\partial_t\boldsymbol{\psi}_v + \partial_{\mathbf{r}}\psi_s + i\partial_{\mathbf{r}} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.49)$$

$$\partial^*\psi = (\partial_t\psi_s - \partial_{\mathbf{r}} \cdot \boldsymbol{\psi}_v) + (\partial_t\boldsymbol{\psi}_v - \partial_{\mathbf{r}}\psi_s - i\partial_{\mathbf{r}} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.50)$$

$$\Phi\psi = (V\psi_s + \mathbf{A} \cdot \boldsymbol{\psi}_v) + (V\boldsymbol{\psi}_v + \psi_s\mathbf{A} + i\mathbf{A} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.51)$$

$$\Phi^*\psi = (V\psi_s - \mathbf{A} \cdot \boldsymbol{\psi}_v) + (V\boldsymbol{\psi}_v - \psi_s\mathbf{A} - i\mathbf{A} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}. \quad (7.52)$$

Writing $F = \mathbf{F} \cdot \boldsymbol{\sigma}$,

$$\text{sc}(iqF\psi) = iq\mathbf{F} \cdot \boldsymbol{\psi}_v, \quad (7.53)$$

$$\text{vec}(iqF\psi) = iq\psi_s\mathbf{F} - q\mathbf{F} \times \boldsymbol{\psi}_v, \quad (7.54)$$

$$\text{sc}(iqF^*\psi) = -iq\mathbf{F}^* \cdot \boldsymbol{\psi}_v, \quad (7.55)$$

$$\text{vec}(iqF^*\psi) = -iq\psi_s\mathbf{F}^* + q\mathbf{F}^* \times \boldsymbol{\psi}_v. \quad (7.56)$$

7.7 Low-energy derivation

For spin zero, write $\Psi' = \exp(-imt)\Psi$. The exact equation becomes

$$(-E^2 - 2mE + \mathbf{P}^2)\Psi = 0 \quad (7.57)$$

$$E = \frac{\mathbf{P}^2 - E^2}{2m} = \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \dots \quad (7.58)$$

The leading term is the Schrödinger equation

$$i\partial_t \Psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \Psi. \quad (7.59)$$

For spin one-half, the two rest-mass-stripped Dirac equations are

$$(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 = 0, \quad (7.60)$$

$$(E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 = 0. \quad (7.61)$$

Eliminating Ψ_1 gives

$$(E^2 + 2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma})\Psi_2 = 0, \quad (7.62)$$

and iteration yields $E = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})/(2m) + O(m^{-2})$, which is equation (7.30).

7.8 Spin probabilities and energy sectors

The measurement projectors act from the left, so the projected amplitudes are $\psi'_{\pm} = \Pi_{\pm}\psi$ as defined above. They are orthogonal and complete:

$$\psi = \psi'_+ + \psi'_- \quad (7.63)$$

$$\langle \psi'_+ | \psi'_- \rangle = 0 \quad (7.64)$$

$$\langle \psi'_- | \psi'_+ \rangle = 0. \quad (7.65)$$

Define the real Bloch vector of a nonzero amplitude by its intrinsic expectation values,

$$s_n := \frac{\text{sc}(\psi^H \sigma_n \psi)}{\text{sc}(\psi^H \psi)} \quad (7.66)$$

$$\mathbf{s} := \{s_1, s_2, s_3\} \quad (7.67)$$

$$\|\mathbf{s}\|^2 \leq 1. \quad (7.68)$$

Equality holds for a pure two-component spin state. The probabilities for spin along and against $\mathbf{u}$ are

$$p_{\pm}(\mathbf{u}) := \frac{\langle \psi'_{\pm} | \psi'_{\pm} \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^H \Pi_{\pm} \Pi_{\pm} \psi)}{\text{sc}(\psi^H \psi)} = \frac{\text{sc}(\psi^H \Pi_{\pm} \psi)}{\text{sc}(\psi^H \psi)} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (7.69)$$

Their conventional column-spinor and density-matrix forms are derived in Appendix A.4.

For a state aligned with $\mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}' = \cos \theta$, the measurement probabilities along $\mathbf{u}'$ are

$$p_+(\mathbf{u}') = \frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2(\theta/2) \quad (7.70)$$

$$p_-(\mathbf{u}') = \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2(\theta/2). \quad (7.71)$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through ϕ about $\mathbf{n}$, then

$$\mathbf{u} \cdot \mathbf{u}' = (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos \phi. \quad (7.72)$$

Thus $\theta = \phi$ when $\mathbf{n} \cdot \mathbf{u} = 0$.

For fluctuations $d\mathbf{u}$ and $d\mathbf{u}'$, the squared determinant identity is

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (7.73)$$

Since the first term is nonnegative,

$$\sqrt{\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}))} \geq \frac{1}{2} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|. \quad (7.74)$$

Finally, on the mass shell let

$$U = \frac{E + \mathbf{P} \cdot \boldsymbol{\sigma}}{m}. \quad (7.75)$$

Since $\det(U) = 1$ and $\mathbf{W}(U)^2 = \mathbf{I}$, the positive- and negative-energy projectors are

$$\frac{1}{2}(\mathbf{I} + \mathbf{W}(U)) \quad (7.76)$$

$$\frac{1}{2}(\mathbf{I} - \mathbf{W}(U)). \quad (7.77)$$

Each squares to itself, their product is zero, and their sum is $\mathbf{I}$. The spin projectors act with the same sign on both components of Ψ , so no additional block notation is needed: their actions are simply $\Pi_+ \Psi$ and $\Pi_- \Psi$. When $\mathbf{W}(U)$ commutes with $\Pi_{\pm}$, as for a compatible spin quantization axis, the four pairwise products are joint energy–spin projectors. They resolve the physical Dirac column selected in Section 7.2 as

$$\Psi = \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\Pi_+ \Psi + \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\Pi_- \Psi \quad (7.78)$$

$$+ \frac{1}{2}(\mathbf{I} - \mathbf{W}(U))\Pi_+ \Psi + \frac{1}{2}(\mathbf{I} - \mathbf{W}(U))\Pi_- \Psi. \quad (7.79)$$

Without this commutation condition the displayed sum remains a sequential decomposition, but the factors are not joint orthogonal projectors.

8 Rotations and Lorentz boosts

8.1 Rotors

A rotation through angle ϕ about unit axis $\mathbf{u}$ is generated by

$$R = \exp\left(\frac{\phi}{2} \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma}\right) = \cos \frac{\phi}{2} + \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma} \sin \frac{\phi}{2} \quad (8.1)$$

$$R^{-1} = R^H. \quad (8.2)$$

The sandwich action $X' = R^H X R$ fixes the scalar component and rotates the spatial vector according to Rodrigues' formula,

$$\mathbf{r}' = \mathbf{r} \cos \phi + (\mathbf{u} \times \mathbf{r}) \sin \phi + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \phi)\mathbf{u}. \quad (8.3)$$

The half-angle explains the spinorial sign change $R(\phi + 2\pi) = -R(\phi)$. The corresponding column-spinor and density-matrix transformations are collected in Appendix A.4. For a spinor $\psi_R := R\psi$, expectation values transform as

$$\psi_R^H = \psi^H R^H \quad (8.4)$$

$$\langle Z \rangle_{\psi_R} = \frac{\text{sc}(\psi^H R^H Z R \psi)}{\text{sc}(\psi^H \psi)} \quad (8.5)$$

$$R^H = R^{-1}. \quad (8.6)$$

For $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$, this gives $Z' = R^H Z R = \mathbf{u}' \cdot \boldsymbol{\sigma}$.

8.2 Boosts

Let rapidity θ satisfy $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. A boost along $\mathbf{u}$ is

$$L = \exp\left(-\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) \quad (8.7)$$

$$L^H = L \quad (8.8)$$

$$L^{-1} = L^*. \quad (8.9)$$

For $\psi_L := L\psi$, the boost generator is Hermitian, so

$$\psi_L^H = \psi^H L \quad (8.10)$$

$$\langle Z \rangle_{\psi_L} = \frac{\text{sc}(\psi^H L Z L \psi)}{\text{sc}(\psi^H L^2 \psi)}. \quad (8.11)$$

The similarity $L^{-1} Z L$ is a different operation: it preserves $\text{sc}(Z)$ and is not the spacetime or spinor-bilinear boost used here. The spacetime paravector therefore transforms by the Hermitian congruence $X' = L X L$, which yields the components $\mathbf{r}_{\parallel} = (\mathbf{r} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$, $\mathbf{r}'_{\parallel} = (\mathbf{r}' \cdot \mathbf{u})\mathbf{u}$, and $\mathbf{r}'_{\perp} = \mathbf{r}' - \mathbf{r}'_{\parallel}$:

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (8.12)$$

$$\mathbf{r}'_{\parallel} = \gamma(\mathbf{r}_{\parallel} - \beta t \mathbf{u}), \quad (8.13)$$

$$\mathbf{r}'_{\perp} = \mathbf{r}_{\perp}. \quad (8.14)$$

It preserves $\det(X) = t^2 - \mathbf{r}^2$. The electromagnetic field instead obeys $F' = L F L^*$ for $F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$. Define $\mathbf{E}_{\parallel} = (\mathbf{E} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}$ and likewise $\mathbf{E}'_{\parallel} = (\mathbf{E}' \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}'_{\perp} = \mathbf{E}' - \mathbf{E}'_{\parallel}$; define the four corresponding components of $\mathbf{B}$ and $\mathbf{B}'$ by the same formulas. Then

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad (8.15)$$

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (8.16)$$

$$\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}, \quad (8.17)$$

$$\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (8.18)$$

With the same convention,

$$\partial' = L \partial L^* \quad (8.19)$$

$$D'^* = L D^* L^*. \quad (8.20)$$

Since L is constant and $L^*L = 1$, the field equation is covariant:

$$\partial' F' = L(\partial F)L^* = LD^*L^* = D'^*. \quad (8.21)$$

Collinear boosts add rapidities and therefore obey

$$\beta' = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2} \quad (8.22)$$

$$\gamma' = \gamma_1\gamma_2(1 + \beta_1\beta_2). \quad (8.23)$$

Inversion changes product order:

$$(LR)^{-1} = R^H L^* \quad (8.24)$$

$$(RL)^{-1} = L^* R^H \quad (8.25)$$

$$(R_1 R_2)^{-1} = R_2^H R_1^H \quad (8.26)$$

$$(L_1 L_2)^{-1} = L_2^* L_1^*. \quad (8.27)$$

9 Generalized spacetime and metric form

Let

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma} \quad (9.1)$$

$$dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}. \quad (9.2)$$

Writing $dX = A dt + \mathbf{B} \cdot d\mathbf{r}$ gives

$$A = \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma} \quad (9.3)$$

$$\mathbf{B} = \partial_r f + \partial_r (\mathbf{g} \cdot \boldsymbol{\sigma}), \quad (9.4)$$

and the interval remains

$$ds^2 = dX dX^* = df^2 - d\mathbf{g}^2. \quad (9.5)$$

For a future-directed time-like curve with $df > 0$,

$$ds = \sqrt{1 - \mathbf{v}^2} df \quad (9.6)$$

$$\mathbf{v}^2 < 1. \quad (9.7)$$

Define the geometric velocity and Lorentz factor by

$$\mathbf{v} = \frac{d\mathbf{g}}{df} \quad (9.8)$$

$$\gamma = \frac{df}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}} \quad (9.9)$$

$$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (9.10)$$

With geometric acceleration $\mathbf{a} = d\mathbf{v}/ds = \gamma d^2\mathbf{g}/df^2$,

$$\frac{dU}{ds} = \gamma^3(\mathbf{v} \cdot \mathbf{a}) + (\gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (9.11)$$

For inertial motion, $dU/ds = 0$. Its scalar component gives $\mathbf{v} \cdot \mathbf{a} = 0$, and its vector component then gives $\mathbf{a} = 0$. Hence $d^2\mathbf{g}/df^2 = 0$, so $\mathbf{g} = c_1f + c_2$ and the geometric velocity is constant. If $f' = df/dt$ and $\mathbf{g}' = d\mathbf{g}/dt$, then

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}''f' - \mathbf{g}'f''}{(f')^2} \quad (9.12)$$

$$[U, dU/ds] = 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \mathbf{i}\boldsymbol{\sigma}. \quad (9.13)$$

A Pauli-matrix representation

A.1 Faithful representation

Every complex element of the displayed algebra basis can be collected as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (A.1)$$

Here $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Such an element is called a complex paravector. Its faithful 2×2 complex representation begins with

$$\text{mat}(\sigma_1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{mat}(\sigma_2) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{mat}(\sigma_3) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix}. \quad (A.2)$$

It preserves identity, addition, and multiplied order:

$$\text{mat}(1) = \mathbf{I}_2, \quad (A.3)$$

$$\text{mat}(Z'Z) = \text{mat}(Z')\text{mat}(Z). \quad (A.4)$$

A.2 Matrix operations and induced algebra operations

Let $\mathbf{M} = \text{mat}(Z)$ with entries $m_{jk} = a_{jk} + b_{jk}i$, where $a_{jk}, b_{jk} \in \mathbb{R}$. Define transpose by

$$(\mathbf{M}^\top)_{jk} := m_{kj}. \quad (A.5)$$

Define entrywise complex conjugation independently by

$$m_{jk}^c := a_{jk} - b_{jk}i \quad (A.6)$$

$$\mathbf{M}^c := \begin{pmatrix} m_{11}^c & m_{12}^c \\ m_{21}^c & m_{22}^c \end{pmatrix}. \quad (A.7)$$

The Hermitian transpose is then

$$\mathbf{M}^H := (\mathbf{M}^c)^\top = (\mathbf{M}^\top)^c. \quad (A.8)$$

Define the algebra operation H through the faithful representation by

$$\text{mat}(Z^H) := \mathbf{M}^H. \quad (A.9)$$

For $n = 1, 2, 3$, its action on every basis type is

$$\text{mat}(1)^{\text{H}} = \mathbf{I}_2, \quad (\text{A.10})$$

$$(\text{i}\mathbf{I}_2)^{\text{H}} = -\text{i}\mathbf{I}_2, \quad (\text{A.11})$$

$$\text{mat}(\sigma_n)^{\text{H}} = \text{mat}(\sigma_n), \quad (\text{A.12})$$

$$\text{mat}(\text{i}\sigma_n)^{\text{H}} = -\text{mat}(\text{i}\sigma_n). \quad (\text{A.13})$$

Hence $1^{\text{H}} = 1$, $\text{i}^{\text{H}} = -\text{i}$, $\sigma_n^{\text{H}} = \sigma_n$, and $(\text{i}\sigma_n)^{\text{H}} = -\text{i}\sigma_n$. Hermitian conjugation changes matrix-product order. Therefore faithfulness gives

$$\text{mat}((\sigma_n \sigma_m)^{\text{H}}) = (\text{mat}(\sigma_n) \text{mat}(\sigma_m))^{\text{H}} \quad (\text{A.14})$$

$$= \text{mat}(\sigma_m)^{\text{H}} \text{mat}(\sigma_n)^{\text{H}} = \text{mat}(\sigma_m \sigma_n), \quad (\text{A.15})$$

and hence

$$(\sigma_n \sigma_m)^{\text{H}} = \sigma_m^{\text{H}} \sigma_n^{\text{H}} = \sigma_m \sigma_n. \quad (\text{A.16})$$

Thus $^{\text{H}}$ is precisely the Hermitian operator defined in Section 2. To define the distinct sigma conjugation, let $\mathbf{S} := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{S}^2 = -\mathbf{I}_2$, so that $\mathbf{S}^{-1} = -\mathbf{S}$, and set

$$\mathbf{M}^* := \mathbf{S}^{-1} \mathbf{M}^{\text{C}} \mathbf{S}, \quad (\text{A.17})$$

$$\text{mat}(\mathbf{Z}^*) := \mathbf{M}^*. \quad (\text{A.18})$$

For $n = 1, 2, 3$, this gives the required action on every basis type:

$$\text{mat}(1)^* = \mathbf{S}^{-1} \mathbf{I}_2^{\text{C}} \mathbf{S} = \mathbf{I}_2, \quad (\text{A.19})$$

$$(\text{i}\mathbf{I}_2)^* = \mathbf{S}^{-1} (\text{i}\mathbf{I}_2)^{\text{C}} \mathbf{S} = -\text{i}\mathbf{I}_2, \quad (\text{A.20})$$

$$\text{mat}(\sigma_n)^* = \mathbf{S}^{-1} \text{mat}(\sigma_n)^{\text{C}} \mathbf{S} = -\text{mat}(\sigma_n), \quad (\text{A.21})$$

$$\text{mat}(\text{i}\sigma_n)^* = \mathbf{S}^{-1} \text{mat}(\text{i}\sigma_n)^{\text{C}} \mathbf{S} = \text{mat}(\text{i}\sigma_n). \quad (\text{A.22})$$

The operation also preserves multiplied order. Since entrywise conjugation preserves order,

$$(\mathbf{M}\mathbf{N})^* = \mathbf{S}^{-1} \mathbf{M}^{\text{C}} \mathbf{N}^{\text{C}} \mathbf{S} \quad (\text{A.23})$$

$$= (\mathbf{S}^{-1} \mathbf{M}^{\text{C}} \mathbf{S}) (\mathbf{S}^{-1} \mathbf{N}^{\text{C}} \mathbf{S}) = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.24})$$

Therefore

$$(\mathbf{M}\mathbf{N})^* = \mathbf{M}^* \mathbf{N}^*. \quad (\text{A.25})$$

The four operations must remain distinct:

symbol	matrix operation	relation to the algebra
$\mathbf{M}^{\text{T}}$	transpose	changes matrix-product order
$\mathbf{M}^{\text{C}}$	entrywise complex conjugation	sends i to $-\text{i}$
$\mathbf{M}^{\text{H}}$	Hermitian transpose	$\text{mat}(\mathbf{Z}^{\text{H}})$
$\mathbf{M}^*$	sigma conjugation	$\text{mat}(\mathbf{Z}^*)$

Their product-order rules are

$$(MN)^\top = N^\top M^\top, \quad (\text{A.26})$$

$$(MN)^{\mathbb{C}} = M^{\mathbb{C}} N^{\mathbb{C}}, \quad (\text{A.27})$$

$$(MN)^{\mathbb{H}} = N^{\mathbb{H}} M^{\mathbb{H}}. \quad (\text{A.28})$$

Sigma conjugation preserves order by equation (A.25). Thus the induced algebra operations satisfy

$$(Z^{\mathbb{H}})^{\mathbb{H}} = Z, \quad (\text{A.29})$$

$$(ZZ')^{\mathbb{H}} = Z'^{\mathbb{H}} Z^{\mathbb{H}}, \quad (\text{A.30})$$

$$(Z^*)^* = Z, \quad (\text{A.31})$$

$$(ZZ')^* = Z^* Z'^*, \quad (\text{A.32})$$

$$Z^{*\mathbb{H}} = Z^{\mathbb{H}*}. \quad (\text{A.33})$$

Trace, determinant, and adjugate. For $\mathbf{M} = \text{mat}(Z)$, define the matrix trace and determinant by

$$\text{tr}(\mathbf{M}) := m_{11} + m_{22} \quad (\text{A.34})$$

$$\det(\mathbf{M}) := m_{11}m_{22} - m_{12}m_{21}. \quad (\text{A.35})$$

The corresponding paravector operations are

$$\text{tr}(Z) := \text{tr}(\mathbf{M}) = 2S, \quad (\text{A.36})$$

$$\det(Z) := \det(\mathbf{M}). \quad (\text{A.37})$$

Faithfulness transfers the matrix identities to the algebra:

$$\text{tr}(Z + Z') = \text{tr}(Z) + \text{tr}(Z'), \quad (\text{A.38})$$

$$\text{tr}(cZ) = c \text{tr}(Z), \quad (\text{A.39})$$

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (\text{A.40})$$

$$\det(ZZ') = \det(Z) \det(Z'), \quad (\text{A.41})$$

$$\det(cZ) = c^2 \det(Z). \quad (\text{A.42})$$

The last identity holds for $c \in \mathbb{C}$. For a 2×2 matrix and its paravector preimage, define

$$\text{adj}(\mathbf{M}) := \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}, \quad (\text{A.43})$$

$$\text{mat}(\text{adj}(Z)) := \text{adj}(\mathbf{M}). \quad (\text{A.44})$$

Direct comparison with the induced algebra operators gives

$$\text{adj}(Z) = Z^{*\mathbb{H}} \quad (\text{A.45})$$

$$\text{mat}(\text{adj}(Z)) = \text{adj}(\mathbf{M}) = (\mathbf{M}^*)^{\mathbb{H}}. \quad (\text{A.46})$$

Equivalently,

$$\text{adj}(Z) = \text{tr}(Z) - Z \quad (\text{A.47})$$

$$\text{tr}(Z) = Z + \text{adj}(Z) = 2S. \quad (\text{A.48})$$

The adjugate changes product order and obeys

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z), \quad (\text{A.49})$$

$$\text{adj}(\text{adj}(Z)) = Z, \quad (\text{A.50})$$

$$\text{adj}(Z) Z = Z \text{adj}(Z) = \det(Z), \quad (\text{A.51})$$

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M}) \mathbf{I}_2. \quad (\text{A.52})$$

Consequently, when $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (\text{A.53})$$

Finally, Cayley–Hamilton gives

$$\mathbf{M}^2 - \text{tr}(\mathbf{M})\mathbf{M} + \det(\mathbf{M})\mathbf{I}_2 = 0, \quad (\text{A.54})$$

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (\text{A.55})$$

A.3 Algebra–matrix crosswalk

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$\text{sc}(Z) := S \quad (\text{A.56})$$

$$\text{vec}(Z) := \mathbf{V}. \quad (\text{A.57})$$

Thus $\text{vec}(Z)$ is the coefficient vector, while its algebra element is $\mathbf{V} \cdot \boldsymbol{\sigma}$.

For $\mathbf{M} = \text{mat}(Z)$, the extraction formulas are

$$\text{sc}(Z) = S = \frac{1}{2} \text{tr}(\mathbf{M}), \quad (\text{A.58})$$

$$\text{mat}(\text{sc}(Z)) = S\mathbf{I}_2, \quad (\text{A.59})$$

$$(\text{vec}(Z))_n = V_n = \frac{1}{2} \text{tr}(\boldsymbol{\sigma}_n \mathbf{M}), \quad (\text{A.60})$$

$$\text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M})\mathbf{I}_2. \quad (\text{A.61})$$

Define real and imaginary paravector extraction through Hermitian splitting:

$$\text{mat}(\text{re}(Z)) := \frac{1}{2}(\mathbf{M} + \mathbf{M}^H), \quad (\text{A.62})$$

$$\text{mat}(\text{im}(Z)) := \frac{\mathbf{M} - \mathbf{M}^H}{2i}. \quad (\text{A.63})$$

By construction, the algebra and matrix operators agree through

$$\text{mat}(Z^H) = \mathbf{M}^H \quad (\text{A.64})$$

$$\text{mat}(Z^*) = \mathbf{M}^* \quad (\text{A.65})$$

$$\text{mat}(Z^{*H}) = (\mathbf{M}^*)^H. \quad (\text{A.66})$$

The matrix-only operations C and T are related to the algebra operations by

$$\mathbf{M}^C = \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1} \quad (\text{A.67})$$

$$\mathbf{M}^T = \mathbf{S} \text{mat}(Z^{*H}) \mathbf{S}^{-1}. \quad (\text{A.68})$$

Finally, scalar extraction of a product and the paravector norm are

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}_{Z'} \mathbf{M}_Z), \quad (\text{A.69})$$

$$\|Z\|^2 := \frac{1}{2} \text{tr}(\mathbf{M}^H \mathbf{M}) = \frac{1}{2} \text{tr}(\mathbf{M} \mathbf{M}^H). \quad (\text{A.70})$$

A.4 Matrix amplitudes and column spinors

This subsection translates the intrinsic paravector spinor development of Section 7.2 into the familiar matrix and column languages. For a paravector amplitude ψ , write

$$\mathbf{M}_\psi := \text{mat}(\psi). \quad (\text{A.71})$$

Faithfulness and Hermitian compatibility give

$$\text{mat}(\psi'^H Z \psi) = \mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_\psi. \quad (\text{A.72})$$

Scalar extraction is half the matrix trace, so the intrinsic definitions (7.1) and (7.2) become

$$\langle \psi' | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_\psi), \quad (\text{A.73})$$

$$\langle \psi' | Z | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_\psi). \quad (\text{A.74})$$

Thus $\text{mat}(\psi^H \psi) = \mathbf{M}_\psi^H \mathbf{M}_\psi$ is generally Hermitian rather than scalar; the scalar inner product is obtained by taking half its trace.

For the fixed unit direction $\mathbf{u}$, the paravector projectors have matrix images

$$\text{mat}(\Pi_\pm) = \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (\text{A.75})$$

Let $\mathbf{v}_+$ be a normalized positive eigenvector,

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \mathbf{v}_+ = \mathbf{v}_+ \quad (\text{A.76})$$

$$\mathbf{v}_+^H \mathbf{v}_+ = 1. \quad (\text{A.77})$$

A convenient phase choice, valid for $u_3 \neq -1$, is

$$\mathbf{v}_+ = \frac{\{1 + u_3, u_1 + iu_2\}^T}{\sqrt{2(1 + u_3)}}. \quad (\text{A.78})$$

At the excluded direction $\mathbf{u} = -\mathbf{e}_3$, use

$$\mathbf{v}_+ = \{0, 1\}^T. \quad (\text{A.79})$$

Then

$$\text{mat}(\Pi_+) = \mathbf{v}_+ \mathbf{v}_+^H. \quad (\text{A.80})$$

For the projected amplitude $\psi' = \Pi_+ \psi$, define

$$\mathbf{M}_{\psi'} := \text{mat}(\psi') = \text{mat}(\Pi_+) \mathbf{M}_\psi \quad (\text{A.81})$$

$$\boldsymbol{\chi} := \mathbf{M}_\psi^H \mathbf{v}_+. \quad (\text{A.82})$$

The algebra, rank-one matrix, and two-component column are related by

$$\boldsymbol{\chi} = \mathbf{M}_{\psi'}^H \mathbf{v}_+ \quad (\text{A.83})$$

$$\mathbf{M}_{\psi'} = \mathbf{v}_+ \boldsymbol{\chi}^H. \quad (\text{A.84})$$

Thus ψ' and $\chi \in \mathbb{C}^2$ contain the same two complex components. The matrix $\mathbf{M}_{\psi'}$ has rank one and has χ^H as its nonzero row precisely when the basis begins with $\mathbf{v}_+$. Conversely, $\mathbf{v}_+ \chi^H$ uniquely determines ψ' . Moreover,

$$\chi^H \chi = \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^H \psi'), \quad (\text{A.85})$$

For an observable acting from the left, the intrinsic matrix element is

$$2 \text{sc}(\psi'^H Z \psi') = \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_{\psi'}) = (\chi^H \chi)(\mathbf{v}_+^H \text{mat}(Z) \mathbf{v}_+). \quad (\text{A.86})$$

The two left-projected matrix amplitudes are

$$\mathbf{M}_{\psi', \pm} := \text{mat}(\Pi_{\pm}) \mathbf{M}_{\psi}. \quad (\text{A.87})$$

Consequently, the intrinsic probabilities in (7.69) have the matrix form

$$p_{\pm}(\mathbf{u}) = \frac{\text{tr}(\mathbf{M}_{\psi}^H \text{mat}(\Pi_{\pm})^H \text{mat}(\Pi_{\pm}) \mathbf{M}_{\psi})}{\text{tr}(\mathbf{M}_{\psi}^H \mathbf{M}_{\psi})} = \frac{\text{tr}(\mathbf{M}_{\psi}^H \text{mat}(\Pi_{\pm}) \mathbf{M}_{\psi})}{\text{tr}(\mathbf{M}_{\psi}^H \mathbf{M}_{\psi})}. \quad (\text{A.88})$$

A.5 Quaternion subalgebra

The real span of the scalar and bivector basis elements is a quaternion subalgebra of the complex paravectors. Define its three imaginary units by

$$q_1 = -\sigma_2 \sigma_3 = -i\sigma_1 \quad (\text{A.89})$$

$$q_2 = -\sigma_3 \sigma_1 = -i\sigma_2 \quad (\text{A.90})$$

$$q_3 = -\sigma_1 \sigma_2 = -i\sigma_3, \quad (\text{A.91})$$

or, collectively, $\mathbf{q} = -i\boldsymbol{\sigma}$. For $n, m \in \{1, 2, 3\}$, the sigma multiplication rule gives

$$q_n q_m = -\delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} q_k, \quad (\text{A.92})$$

Consequently,

$$q_1^2 = q_2^2 = q_3^2 = -1 \quad (\text{A.93})$$

$$q_1 q_2 = q_3 \quad (\text{A.94})$$

$$q_1 q_2 q_3 = -1, \quad (\text{A.95})$$

with the remaining products fixed by anticommutation. For $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$, define the real paravector

$$Q = a + \mathbf{A} \cdot \mathbf{q} = a - i\mathbf{A} \cdot \boldsymbol{\sigma}. \quad (\text{A.96})$$

These are precisely the quaternions embedded in the paravector algebra. For a second quaternion $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$, equation (A.92) gives the familiar product

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}. \quad (\text{A.97})$$

Hermitian conjugation reverses every bivector, while sigma conjugation fixes it:

$$q_n^H = -q_n \quad (\text{A.98})$$

$$q_n^* = q_n. \quad (\text{A.99})$$

Quaternion conjugation is therefore already the Hermitian operation,

$$Q^H = a - \mathbf{A} \cdot \mathbf{q} \quad (\text{A.100})$$

$$Q^* = Q \quad (\text{A.101})$$

$$\text{adj}(Q) = Q^H. \quad (\text{A.102})$$

It follows that

$$QQ^H = Q^H Q = a^2 + \mathbf{A}^2 = \det(Q) \quad (\text{A.103})$$

$$Q^{-1} = \frac{Q^H}{a^2 + \mathbf{A}^2} \quad (\text{A.104})$$

for nonzero Q .

The faithful matrix representation sends the quaternion units to

$$\text{mat}(q_1) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \quad (\text{A.105})$$

$$\text{mat}(q_2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (\text{A.106})$$

$$\text{mat}(q_3) = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}. \quad (\text{A.107})$$

Hence equation (A.96) becomes

$$\text{mat}(Q) = a\mathbf{I}_2 - i\mathbf{A} \cdot \boldsymbol{\sigma} = \begin{pmatrix} a - iA_3 & -A_2 - iA_1 \\ A_2 - iA_1 & a + iA_3 \end{pmatrix}. \quad (\text{A.108})$$

The algebraic conjugation and norm become the usual matrix identities

$$\text{mat}(Q^H) = \text{mat}(Q)^H, \quad (\text{A.109})$$

$$\text{mat}(Q)^H \text{mat}(Q) = \text{mat}(Q) \text{mat}(Q)^H = (a^2 + \mathbf{A}^2)\mathbf{I}_2, \quad (\text{A.110})$$

$$\det(\text{mat}(Q)) = a^2 + \mathbf{A}^2. \quad (\text{A.111})$$

Thus a unit quaternion is represented by a unitary matrix of determinant one.

Finally, for a unit axis $\mathbf{u}$, let

$$Q = \cos \frac{\theta}{2} + \mathbf{u} \cdot \mathbf{q} \sin \frac{\theta}{2}. \quad (\text{A.112})$$

Since $\mathbf{q} = -i\boldsymbol{\sigma}$, this quaternion is R^H for the rotor R defined in Section 8. The quaternion sandwich

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H \quad (\text{A.113})$$

therefore reproduces the same Rodrigues rotation.

B Reference summary

This index collects the principal definitions and results in natural units, with signature $(+, -, -, -)$. Topic titles and equation numbers link to their full statements and derivations.

Algebra and vector operations.

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9),$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9),$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma},$$

$$\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A},$$

$$\|\mathbf{Z}\|^2 = \|\mathbf{S}\|^2 + \|\mathbf{V}\|^2,$$

$$\|\mathbf{V}\|^2 = \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} \quad (3.83).$$

Hermitian and sigma-conjugate operators.

$$(ZZ')^* = Z^* Z'^*,$$

$$(ZZ')^H = Z'^H Z^H,$$

$$\text{adj}(Z) = Z^{*H},$$

$$\text{adj}(ZZ') = Z'^{*H} Z^{*H} \quad (3.13).$$

Paravectors, extraction, and products. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + B\mathbf{i}) \cdot \boldsymbol{\sigma}$,

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}),$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*H}),$$

$$a = \text{sc}(\text{re}(Z)),$$

$$b = \text{sc}(\text{im}(Z)),$$

$$\mathbf{A} = \text{vec}(\text{re}(Z)),$$

$$\mathbf{B} = \text{vec}(\text{im}(Z)),$$

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma},$$

$$\det(Z) = \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2,$$

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)},$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0,$$

$$\lambda_{\pm} = S \pm \sqrt{\mathbf{V}^2}.$$

See the product (3.68), determinant (3.15), and quadratic identity (3.55).

Projection operators and functions.

$$\begin{aligned}
\Pi_{\pm} &= \tfrac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
\Pi_{\pm}^2 &= \Pi_{\pm}, \\
\Pi_+ \Pi_- &= 0, \\
f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_-, \\
X &= \lambda_+ \Pi_+ + \lambda_- \Pi_-, \\
f(X) &= f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-.
\end{aligned}$$

See the general complex-paravector rule (3.160) and ladder operators in Section 3.7.

Space–time kinematics and action.

$$\begin{aligned}
dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\
ds^2 &= dX dX^* = dt^2 - d\mathbf{r}^2 \tag{5.3}, \\
U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
\gamma &= (1 - \mathbf{v}^2)^{-1/2}, \\
mU &= E + \mathbf{P} \cdot \boldsymbol{\sigma}, \\
E^2 - \mathbf{P}^2 &= m^2 \tag{5.21}, \\
\int L dt &= -m \int ds, \\
L &= -m/\gamma, \\
H &= m\gamma, \\
K &= H - m = m(\gamma - 1) \tag{5.54}.
\end{aligned}$$

Electromagnetism.

$$\begin{aligned}
\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
F &= (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, \\
D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\
\partial F &= D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma} \tag{6.4}, \\
\Box &= \partial^* \partial, \\
\partial \Phi &= S + F^*, \\
\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\
\mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\
\rho &= \Box V - \partial_t S \\
\mathbf{J} &= \Box \mathbf{A} + \partial_{\mathbf{r}} S, \\
\tfrac{1}{2}F^2 &= \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + \mathbf{i} \mathbf{E} \cdot \mathbf{B}, \\
\tfrac{1}{2}F F^{\mathbf{H}} &= \tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
m \frac{dU}{ds} &= q \operatorname{re}(FU), \\
\frac{d\mathbf{P}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{6.50}.
\end{aligned}$$

See the field invariants (6.46) and Poynting balance (6.82).

Wave equations, spin, and energy sectors.

$$\begin{aligned}
\langle \psi' | \psi \rangle &= \text{sc}(\psi'^{\mathbf{H}} \psi), \\
\langle \psi' | Z | \psi \rangle &= \text{sc}(\psi'^{\mathbf{H}} Z \psi), \\
\langle Z \rangle_{\psi} &= \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^{\mathbf{H}} Z \psi)}{\text{sc}(\psi^{\mathbf{H}} \psi)}, \\
(\square + m^2)\Psi &= 0, \\
\mathbf{W}(Z) &= \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}, \\
(\mathbf{W}(\text{i}\partial) - m\mathbf{I})\Psi &= 0, \\
(\mathbf{W}(\text{i}\partial - q\Phi^*) - m\mathbf{I})\Psi &= 0, \\
\text{i}\partial_t \Psi &= \left(\frac{(-\text{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi, \\
\Pi_{\pm} &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
p_{\pm}(\mathbf{u}) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}), \\
\text{positive-energy projector:} & \quad \frac{1}{2}(\mathbf{I} + \mathbf{W}(U)), \\
\text{negative-energy projector:} & \quad \frac{1}{2}(\mathbf{I} - \mathbf{W}(U)), \\
\text{positive-spin projection:} & \quad \Pi_+ \Psi, \\
\text{negative-spin projection:} & \quad \Pi_- \Psi.
\end{aligned}$$

See the paravector inner product (7.1), minimal-ideal development in Section 7.2, Klein–Gordon equation (7.16), Dirac equations (7.24) and (7.29), general Klein–Gordon lift (7.28), Pauli equation (7.30), and spin probability (7.69).

Rotations and boosts.

$$\begin{aligned}
R &= \exp(\phi \mathbf{u} \cdot \text{i}\boldsymbol{\sigma}/2), \\
R^{\mathbf{H}} &= R^{-1}, \\
Z' &= R^{\mathbf{H}} Z R, \\
L &= \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \\
L^{\mathbf{H}} &= L, \\
X' &= L X L, \\
\langle Z \rangle_{\psi_T} &= \frac{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} Z T \psi)}{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} T \psi)}.
\end{aligned}$$

See the bilinear rule (7.4); varying-frame derivatives are developed in Section 5.2.

Matrix representation in Appendix A.

$$\text{mat}(Z) = \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix} \quad (\text{A.2}),$$

$$\text{mat}(ZZ') = \text{mat}(Z) \text{mat}(Z') \quad (\text{A.4}),$$

$$\text{mat}(Z^H) = \text{mat}(Z)^H,$$

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \text{mat}(Z)^c \mathbf{S} \quad (\text{A.18}),$$

$$\text{mat}(\text{adj}(Z)) = \text{adj}(\text{mat}(Z)),$$

$$\det(Z) = \det(\text{mat}(Z)) \quad (\text{A.52}).$$

The minimal-ideal column map is equation (A.83).

Quaternion subalgebra in Appendix A.

$$\begin{aligned} \mathbf{q} &= -i\boldsymbol{\sigma}, \\ Q^H &= a - \mathbf{A} \cdot \mathbf{q}, \\ QQ^H &= a^2 + \mathbf{A}^2. \end{aligned}$$

See the quaternion product (A.92), its paravector embedding (A.96), and its faithful matrix form (A.108).

Generalized spacetime and metric form.

$$\begin{aligned} X &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, \\ ds^2 &= df^2 - d\mathbf{g}^2, \\ \mathbf{v} &= d\mathbf{g}/df, \\ U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

C Conventions and cautions

- The same symbol $i = \sigma_1\sigma_2\sigma_3$ is both the algebra's pseudoscalar and, in the Pauli representation, the ordinary complex unit.
- Sigma conjugation $(\cdot)^*$, the Hermitian operator $(\cdot)^H$, and their composition are distinct. In the Pauli representation, the Hermitian operator corresponds to Hermitian adjunction.
- $\det(Z)$ is an indefinite quadratic form and may be complex for complex Z .
- Differential operators act on everything to their right unless parentheses state otherwise. This matters in products involving fields.
- Vectors and matrices are bold, and only vectors and matrices are bold. In particular, the electromagnetic fields formerly written as calligraphic or italic E and B are the upright bold vectors $\mathbf{E}$ and $\mathbf{B}$, while scalar energy remains the unbold symbol E . Blackboard bold is retained only for the number sets $\mathbb{R}$ and $\mathbb{C}$.
- The action is written as $\int L dt$, avoiding conflict with the scalar part S . The Lagrangian L and Hamiltonian H use the regular math font. The Dirac block function is the upright bold $\mathbf{W}(\cdot)$, and a vector square is explicitly defined by $\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}$.

- Curly braces are reserved for explicit vector-entry lists, and commutators alone use square brackets. Parentheses delimit matrices, functions, algebraic grouping, and order-sensitive operator products. Redundant grouping parentheses were removed.
- The extraction functions $\text{sc}(\cdot)$, $\text{vec}(\cdot)$, $\text{re}(\cdot)$, and $\text{im}(\cdot)$ replace **scalar**, **vector**, and their capitalized variants. They are defined before use. The unused expectation abbreviation **Exp** was removed.
- The original metric notation $\langle\langle Z \rangle\rangle^2$ and determinant bars were replaced by $\det(Z)$. Double bars are reserved for squared norms, with

$$\|Z\|^2 = \|S\|^2 + \|\mathbf{V}\|^2.$$

Absolute-value bars are not used for paravector norms. Entrywise complex-conjugation notation is confined to the matrix appendix.

- Sigma conjugation retains the splat $*$. The operation originally written R , and elsewhere as a dagger, is written H because it reverses multiplied sigma order and becomes Hermitian adjunction in the matrix representation. The abbreviation $\text{adj}(Z) := Z^{*H}$ is defined early and used where it shortens formulas.
- The matrix appendix uses matching upright sans-serif capitals T , C , and H for transpose, entrywise complex conjugation, and Hermitian transpose. It distinguishes these operations from sigma conjugation and states which operations reverse product order.
- Plus/minus labels are subscripts ($\pm, +, -$), not superscripts or parenthesized labels. Parallel and perpendicular components use $\parallel$ and $\perp$ rather than **par** and **prp**; their projection formulas are restated where needed.
- Spatial derivatives use ∂_r rather than ∇ , and no overbars are used. Literal or malformed source commands such as **quad** and **frac** were converted to formatted L^AT_EX.
- Long product expansions were consolidated while retaining their distinct components and results. Expressions otherwise preserve the compact ordering of the source, for example $\text{sc}(\Phi^*U)$, without an extra pair of grouping parentheses.